
Mapping Cross-Model Conflict in Protein Fitness Prediction: Assay-Level Structure, Variant-Level Noise, and the Null Effect of Chain-of-Thought Arbitration

NeuriCo

Abstract

Protein engineering campaigns are budget-limited: a laboratory synthesises tens to hundreds of designed variants, and which variants make that list is decided by computational fitness predictors. There are now roughly 95 such predictors and no agreed way to choose among them, so when they disagree the standard practice is to average and hope. We ask whether that disagreement is itself a usable signal. We build a family-balanced panel of 30 protein foundation models drawn from 10 independently-developed lineages, measure their conflict on 216 deep mutational scanning assays and 2.47M variants, decompose it into within-family (aleatoric) and cross-family (epistemic) components, and test whether surfacing it to a chain-of-thought arbiter improves ensemble weighting. Conflict is strongly structured at the assay level: dataset covariates predict it with held-out $R^2 = 0.414$, it is highest for shallow alignments and viral proteins, and cross-family disagreement flags confident ensemble failures that within-family agreement hides ($\Delta\rho = 0.108$, $d = 0.95$). It is almost unstructured at the variant level, where every conflict signal detects errors at AUROC ≈ 0.51 . Showing the conflict to an LLM doubles the correlation between its assigned weights and the models' true accuracy ($0.235 \rightarrow 0.477$), but chain-of-thought over the same evidence adds nothing ($\Delta\rho = -0.0003$, $p = 0.49$), the end-task gain loses significance once option order is randomised, and a 40-line ridge regression given the same inputs captures an order of magnitude more of the available headroom. Use conflict as an assay-level trust score and a routing signal, not as a per-variant filter.

1 Introduction

A protein engineering campaign is a list. A laboratory can synthesise and assay tens to hundreds of designed variants, not millions, and which variants make that list is decided by computational fitness predictors. There are now about 95 published zero-shot predictors on the PROTEINGYM benchmark [Notin et al., 2023] and no agreed way to choose among them. When they disagree—and they disagree constantly—the standard practice is to average their scores and hope.

We ask whether the disagreement itself is worth reading. The hypothesis we test is that conflict between protein foundation models is not noise but signal: that it clusters around specific biological ambiguities and dataset biases, that it therefore marks where an ensemble should not be trusted, and that surfacing it through chain-of-thought reasoning lets a language model arbitrate between predictors and improve variant selection. This is an economically meaningful question. If disagreement is usable, it saves wet-lab budget by flagging untrustworthy regions before synthesis; if it is not, that is worth knowing before more effort is spent building conflict-based arbitration schemes.

The gap. Disagreement is already exploited in this field, but never characterised. Tan et al. [2026] have an LLM arbitrate between six protein predictors, weighting them *unconditionally*, and use the union of each model’s top candidates to widen recall—so conflict is a nuisance parameter, never an object of study. No published work maps cross-model conflict onto PROTEINGYM’s biological covariates. Separately, Hamidieh et al. [2026] split uncertainty into within-source noise (aleatoric, AU) and cross-source disagreement (epistemic, EU) for LLM self-consistency, and make the sharp prediction that *low AU with high EU marks confident failures*; that prediction has never been tested where the sources are genuinely independent research lineages rather than samples from one model. Finally, the chain-of-thought literature’s own correctives—that debate and CoT gains routinely vanish against compute-matched controls [Zhang et al., 2025b, Smit et al., 2023]—are almost never applied in protein-LLM papers, and no one separates whether the reasoning identified the *right* biological ambiguity from whether the score went up.

Our approach. We build a family-balanced panel of 30 models drawn from 10 independently-developed lineages of PROTEINGYM’s score matrix (section 3), so that no single lineage can dominate either the consensus or the conflict estimate. Because the panel reuses published per-variant scores, the entire conflict analysis runs with zero protein model inference: the models are the subject of study, not a dependency. We decompose per-variant conflict into AU and EU, regress it on biological and dataset covariates, test whether it detects ensemble failure, and then run a 2×2 ablation—conflict shown or hidden, crossed with chain-of-thought or direct answer—on two locally-run open-weight arbiters, against a full baseline ladder that includes a compute-matched CoT control, an order-randomised control, a cheating oracle, and a non-LLM ridge meta-weighter with access to exactly the same information.

What we find. Conflict is strongly structured, but only at the level of assays and strata. Dataset covariates predict an assay’s mean cross-family disagreement with 5-fold CV $R^2 = 0.414$ (permutation $p = 0.005$); disagreement is highest where alignments are shallow ($\rho = -0.43$ with MSA depth) and on viral proteins (27% above human); and within the low-AU stratum, where a self-consistency estimate would report confidence, high cross-family disagreement costs $\Delta\rho = 0.108$ of ensemble accuracy (Cohen’s $d = 0.95$). At the level of individual variants the same signal nearly vanishes: per-assay AUROC for detecting the worst ensemble errors is 0.511. Surfacing the conflict to an LLM robustly doubles the quality of its reasoning about which models to trust (weight-accuracy correlation $0.235 \rightarrow 0.477$ under an order-randomised design), yet moves the end metric by only $+0.0022$ Spearman, which is not significant after correction—while a ridge regression on the same inputs gains $+0.0336$, or 61% of the entire oracle headroom.

- We produce the first quantitative cartography of cross-model conflict on PROTEINGYM: 30 models in 10 lineages, 216 assays, 2.47M variants, with conflict decomposed into aleatoric and epistemic components and regressed on biological and dataset covariates with permutation tests and held-out validation.
- We transplant the confident-failure prediction of Hamidieh et al. [2026] out of LLM self-consistency and into protein foundation models, where it replicates with a large effect ($d = 0.95$).
- We run a 2×2 ablation that separates *showing* the conflict from *reasoning about* it, with the compute-matched CoT control the debate literature demands, an order-randomised control, two backbones, three seeds, and a non-LLM competitor—and show that the evidence does the work while the reasoning format does not.
- We measure rationale correctness separately from end-task gain, and expose a systematic arbiter failure mode: it reads “this family is an outlier” as “this family is wrong”, and therefore under-trusts inverse-folding models precisely where they are most accurate.

section 2 positions the work, section 3 describes the panel and the experiments, section 4 reports results, and section 5 draws out what practitioners should and should not do with cross-model conflict.

2 Related Work

Zero-shot protein fitness prediction and ensembling. Zero-shot variant effect predictors score mutations without labels, and fall into several genuinely distinct lineages: Potts and VAE models over multiple sequence alignments [Hopf et al., 2017, Riesselman et al., 2018, Frazer et al., 2021], evolutionary conservation methods [Laine et al., 2019], protein language models trained with

masked or autoregressive objectives [Meier et al., 2021, Lin et al., 2023, Nijkamp et al., 2023], alignment-conditioned generative models [Notin et al., 2022, Truong Jr and Bepler, 2023], and structure-conditioned or inverse-folding models [Hsu et al., 2022, Dauparas et al., 2022, Su et al., 2024, Li et al., 2024]. PROTEINGYM [Notin et al., 2023] collects 217 deep mutational scanning assays and, crucially for us, *publishes the per-variant scores of roughly 95 such models*, which makes cross-model conflict directly measurable without re-running anything. The dominant way to combine them is uniform averaging. Tan et al. [2026] go further, using an LLM to set ensemble weights from taxonomy, alignment depth and structural context, followed by a chain-of-thought audit panel. Unlike that work, which weights predictors unconditionally and never asks where they conflict, we condition explicitly on measured disagreement and we test whether conditioning helps at all.

Disagreement as uncertainty. In the LLM literature, Hamidieh et al. [2026] formalise a black-box split between aleatoric uncertainty (dispersion of a model’s own resampled responses) and epistemic uncertainty (the gap between a model’s self-similarity and its similarity to independently-trained peers), and argue that cross-model disagreement is highest on incorrect answers precisely when aleatoric uncertainty is low. Their ensemble design criteria—support richness, non-collapsing diversity, calibrated weighting—are what force our family-balanced panel: averaging PROTEINGYM’s raw 95 columns would be dominated by whichever lineage contributes the most members, silently converting cross-lineage disagreement into within-lineage scale noise. Within-model ensembling for protein models exists in the form of inference-time dropout [Jiang et al., 2025], which is the aleatoric-side analogue. Building on Hamidieh et al. [2026], we transplant the decomposition to a setting where the “sources” are separate research programmes rather than samples from one model.

Chain-of-thought, debate, and their correctives. Chain-of-thought prompting [Wei et al., 2022] is the instrument the hypothesis names. The multi-agent literature has since produced its own correctives: Smit et al. [2023] and Zhang et al. [2025b] both report that debate and CoT gains frequently disappear once the single-model baseline is matched on test-time compute, and that what actually helps is heterogeneity among the participants. Zhang et al. [2025c] document degradation as agents homogenise, and Lee et al. [2025] find that reasoning can actively hurt a model’s ability to represent genuine disagreement. Following these, every LLM condition we report is paired with a compute-matched control that reasons for the same token budget without seeing the conflict. Xu et al. [2024] survey conflict for LLMs and supply the vocabulary; our conflicts are between external predictors rather than between context and parametric memory.

Failure-mode and slice discovery. A mature methodology exists for finding coherent, describable subpopulations where a model underperforms [Eyuboglu et al., 2022, Chen et al., 2025], and the standard it sets is that a discovered slice must be predictive on held-out data, not merely a low-performing bucket found post hoc. It has not been applied to protein foundation models. We adopt the standard—our slice discovery is validated on held-out assays—and report honestly that it only partially succeeds.

Dataset bias. The “dataset biases” half of the hypothesis has direct support: protein databases encode survivorship bias [Shaw et al., 2026], and in the genomic domain benchmark construction alone moves reported performance [Zhang et al., 2025a] and cross-model comparison is systematically confounded [Dalla-Torre et al., 2026]. The last two are why we instantiate the hypothesis on protein rather than genomic foundation models (section 3): no genomic benchmark publishes a comparable cross-model per-example score matrix, so measured “disagreement” there would be inseparable from evaluation-protocol artefacts.

3 Method and Experimental Setup

3.1 Data

We use PROTEINGYM v1.3, substitution benchmark [Notin et al., 2023]: 217 deep mutational scanning assays with experimental fitness measurements, 95 published zero-shot model score columns per assay, and 46 metadata covariates per assay. We retain 216 assays and **2,465,704 variants**, dropping NPC1_HUMAN_Erwood_2022_RPE1 for incomplete model coverage. Of the retained variants, 696,248 are single substitutions, which is the subset used wherever mutation-level features are required. Because we reuse the published score matrix, the entire conflict analysis runs with zero protein model inference.

FAMILY	MODELLING PARADIGM	MEMBERS (3 PER FAMILY)	MEAN CENTRALITY RANK
MSA-POTTS	Potts / VAE over MSAs	EVmutation, DeepSequence-ens, EVE-ens	7.32
Evo_Conservation	Evolutionary conservation	GEMME, ESCOTT, VESPA	5.26
MSA-CONDITIONED	Alignment-conditioned generative	MSA-Transformer-ens, PoET, TranceptEVE-L	1.65
ESM_Masked	Masked protein LM	ESM-1v-ens, ESM2-650M, ESM-C-600M	3.48
AR_PLM	Autoregressive protein LM	ProGen2-large, RITA-L, ProGen3-1B	7.36
xTrimoPGLM	Large hybrid MLM/CLM	xTrimoPGLM-3B-MLM, -3B-CLM, -10B-MLM	7.17
INVERSE-FOLDING	Structure-conditioned design	ESM-IF1, ProteinMPNN, MIF-ST	9.51
Struct_PLM	Structure-aware PLM	ProSST-2048, ProtSSN-ens, SaProt-650M-AF2	4.62
Struct_MSA_Hybrid	Structure + retrieval hybrid	VENUSREM, S3F-MSA, RSALOR	3.32
CNN_RNN	Convolutional / recurrent	CARP-640M, UniRep-evotune, WaveNet	5.32

Table 1: The family-balanced panel: 10 independently-developed lineages, 3 members each. Balancing is required because PROTEINGYM’s raw 95 columns are dominated by a few lineages (ProtSSN contributes 10 columns, ProSST 6, ESM2 6), which would reclassify within-lineage scale noise as cross-lineage disagreement. Centrality rank is the family’s mean rank by agreement with the other nine (1 = most central); INVERSE-FOLDING is the most distinctive lineage in the panel, which matters in section 4.6.

A flagged reinterpretation. The hypothesis we test names *genomic* foundation models; we instantiate it on *protein* foundation models. The reason is a decisive feasibility asymmetry. No genomic benchmark publishes a per-example cross-model score matrix comparable to PROTEINGYM’s, so every DNA model would have to be re-run from scratch, and Dalla-Torre et al. [2026] and Zhang et al. [2025a] document that genomic models are currently not comparable across benchmarks—measured “disagreement” would be confounded with evaluation-protocol artefacts. Protein foundation models are the sequence-model family where a clean test is possible today. A genomic replication is the natural follow-up once such a matrix exists.

3.2 The family-balanced panel

PROTEINGYM’s 95 columns are not 95 independent opinions. We group them into 10 independently-developed lineages and take exactly 3 members from each, giving 30 models in which every family contributes equally to both the ensemble and the conflict estimate (table 1). This follows the non-collapsing-diversity criterion of Hamidieh et al. [2026]: if one lineage dominated the panel, within-lineage scale noise would masquerade as cross-lineage conflict. Model outputs live on incomparable scales, so every model’s scores are converted to within-assay normalised ranks in $[0, 1]$ before any aggregation.

3.3 Conflict decomposition

Let $\mathcal{F} = \{f_1, \dots, f_{10}\}$ be the families, each with members $m \in f$, and let $r_m(v) \in [0, 1]$ be model m ’s normalised rank prediction for variant v . Following Hamidieh et al. [2026], we define for each variant

$$AU(v) = \frac{1}{|\mathcal{F}|} \sum_{f \in \mathcal{F}} \text{sd}_{m \in f} r_m(v), \quad \bar{r}_f(v) = \frac{1}{|f|} \sum_{m \in f} r_m(v), \quad (1)$$

$$EU(v) = \text{sd}_{f \in \mathcal{F}} \bar{r}_f(v), \quad TU(v) = AU(v) + EU(v), \quad (2)$$

with the family-balanced ensemble prediction $\hat{y}(v) = \frac{1}{|\mathcal{F}|} \sum_f \bar{r}_f(v)$. AU is the aleatoric analogue—dispersion *within* a modelling paradigm—and EU the epistemic analogue—dispersion *across* paradigms. Assay-level conflict is the mean of the per-variant quantity over that assay’s variants.

3.4 Experiments

E1: conflict cartography. We regress assay-level and mutant-level EU on PROTEINGYM covariates (alignment depth N_{eff}/L , number of homologous sequences, MSA coverage, sequence length, taxon, coarse selection type, region mutated, fraction of multi-mutants) and on mutation-level chemistry (BLOSUM62 score, $|\Delta \text{ hydrophobicity}|$, $|\Delta \text{ volume}|$, $\Delta \text{ charge}$, relative position). Associations use 2000-draw permutation tests with Benjamini–Hochberg FDR; multivariate models are validated with 5-fold CV, and the mutant-level gradient-boosted model uses GroupKFold over assays so that it is scored on held-out proteins.

E2: failure detection. We rank variants by conflict, abstain on the most-conflicted fraction, and report selective Spearman against random, confidence-magnitude, and AU-only abstention. We then

test the confident-failure prediction directly by splitting each assay at its median AU and median EU and comparing ensemble accuracy across the four cells. Slice discovery clusters conflict on 60% of assays and validates the clusters on the held-out 40%.

E2b: fair detector comparison. Selective Spearman is confounded by range restriction—abstaining on mid-range predictions inflates rank correlation whether or not the abstained points were wrong. We therefore re-rank the detectors by per-assay AUROC for identifying the worst 20% of ensemble errors, a metric with no such artefact, and additionally fit a held-out-assay logistic combination of the signals.

E3: conflict-driven CoT arbitration. The arbiter allocates integer weights 0–10 across the 10 families *for one assay*, given the protein and assay context and, in the conflict conditions, a numerical profile of how the families disagree on that assay. The weighted mean of family rank predictions is scored against the experimental data. Aiming the arbiter at the assay level rather than the variant level is a deliberate consequence of E1 and E2: that is where the conflict signal lives. The design is a 2×2 :

	CONFLICT PROFILE SHOWN	CONFLICT PROFILE HIDDEN
chain-of-thought	CONFLICT-CoT (proposed)	CoT-ONLY (compute-matched control)
direct answer	CONFLICT-DIRECT	DIRECT-ONLY (metadata-only control)

E4: rationale correctness. We correlate the 10 assigned weights against the families’ *true* per-assay Spearman (which the model never sees), probe the generated text for whether it names the biological factor actually present in the assay metadata, check whether the weights merely follow agreement-centrality, and rank families by assigned weight versus true accuracy to expose systematic mis-ranking.

E5: design selection. The candidate pool is the ensemble’s top-100 variants per assay; a budget of K variants is “synthesised”; the metric is normalised maximum true fitness among the K selected (0 = worst variant in the assay, 1 = best) plus hit rate on the true top 1%, over 207 assays. We test conflict as a *filter* (penalising high-conflict candidates with strength λ) and as a *router* (a partial oracle that perfectly resolves one half of the pool, with random-half and low-conflict-half controls).

E6/E7: robustness. E6 measures positional bias in the option list, backbone agreement, seed stochasticity, and prompt-length confounds. E7 replicates the primary contrast with the family list permuted per assay.

3.5 The LLM arbiter

No LLM API key was available in this environment, so all chain-of-thought is generated by open-weight models run locally: QWEN3-14B (bf16) as the primary reasoner and QWEN3-8B (bf16) as a backbone ablation. Nothing is simulated. Decoding is greedy for the primary runs and temperature 0.7 / top- p 0.9 for two extra sampled seeds of CONFLICT-CoT and CoT-ONLY on the 14B backbone. We set `enable_thinking=False`, so the reasoning that appears is the reasoning we asked for rather than an internal trace, and cap generation at 600 new tokens for CoT conditions and 96 for direct conditions. In total we generate **2,808** completions across 13 run configurations, of which 2,807 parse to a complete 10-family weight vector.

Leakage control. Every number in every prompt derives from model predictions and dataset meta-data only. Experimental fitness scores and per-assay model accuracies never enter a prompt; we verified this programmatically over all 1,296 prompts (0 hits for any ground-truth keyword).

3.6 Baselines

The ladder is: BEST-SINGLE (VENUSREM, the strongest individual model in the panel); UNIFORM, the family-balanced uniform ensemble, which is *the bar to beat*; RANDOM-WEIGHTS (20 draws); LEARNED-META, a ridge regression predicting each family’s per-assay accuracy from the same covariates and conflict statistics the LLM sees, trained on held-out assays via GroupK-Fold, plus its `noconflict` and `conflictonly` ablations; the four LLM conditions and their order-randomised variants; and ORACLE, whose weights are fitted on the same assay’s ground truth—a cheating upper bound that quantifies how much headroom per-assay reweighting has at all.

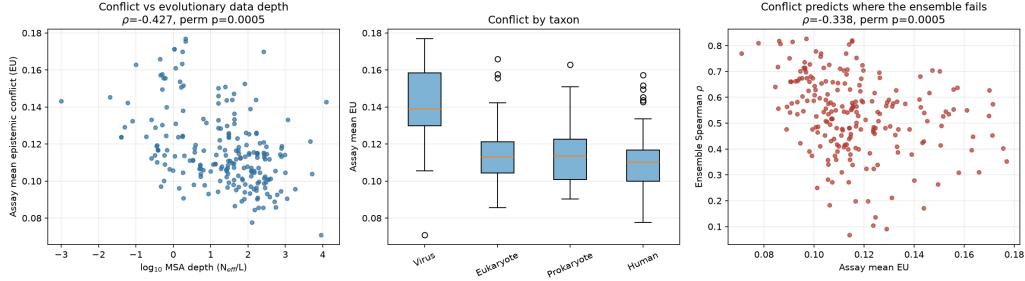


Figure 1: **Cross-model conflict is structured at the assay level.** (Left) An assay’s mean cross-family conflict (EU) falls with the depth of its evolutionary data ($\rho = -0.427$, permutation $p = 0.0005$). (Center) Viral proteins carry 27% more conflict than human proteins. (Right) Conflict tracks where the ensemble is wrong: assays with more disagreement have lower ensemble accuracy ($\rho = -0.338$, permutation $p = 0.0005$).

CONTINUOUS COVARIATE	ρ with EU	perm p	FDR p	CATEGORICAL	KW p	η^2	Ordering (mean EU)
$\log_{10}$ # homologous sequences	-0.452	0.0005	0.0010	taxon	0.0005	0.181	Virus > Euk. > Prok. > Human
$\log_{10}$ MSA depth (N_{eff}/L)	-0.427	0.0005	0.0010	MSA depth cat.	0.0005	0.138	Low > Medium > High
MSA coverage %	+0.244	0.0010	0.0016	selection type	0.032	0.031	Binding > Fitness > Activity
$\log_{10}$ sequence length	+0.228	0.0015	0.0022	region mutated	0.11	0.010	n.s.
fraction multi-mutants	-0.208	0.0005	0.0010				
$\log_{10}$ # variants	-0.021	0.77	0.77				

Table 2: **Assay-level conflict is predicted by dataset covariates.** (Left) Spearman correlation of each continuous covariate with an assay’s mean cross-family conflict (EU), with 2000-draw permutation p -values and BH-FDR correction. (Right) Kruskal–Wallis tests for categorical covariates. Conflict concentrates exactly where the biology is hardest to constrain: shallow alignments, viral proteins (mean EU 0.141 vs 0.111 for human, +27%), and binding assays. Strongest association in each panel in **bold**.

3.7 Metrics and statistics

The primary metric is mean per-assay Spearman ρ between predicted and experimental fitness; we also report ρ_{PG} , PROTEINGYM’s official bias-corrected aggregation, which averages within (UniProt ID, selection type) groups first. Assays are the unit of analysis throughout. Paired comparisons use the Wilcoxon signed-rank test over assays with Cohen’s d on the paired differences; confidence intervals are 2000-draw bootstraps over assays; covariate associations use 2000-draw permutation tests with BH-FDR; the arbitration ladder is Holm-corrected across methods. Mutant-level correlations are reported with cluster-robust inference (a per-assay statistic, then a one-sample t -test over assays), because pooling 2.5M variants across 216 assays would give meaninglessly small p -values. We fix SEED = 42; the environment is Python 3.12.8, numpy 2.2.6, pandas 3.0.5, scipy 1.18.0, scikit-learn 1.9.0, torch 2.13.0+cu130, transformers 5.15.0, on $2 \times$ NVIDIA RTX A6000 (48 GB), with a total wall-clock of roughly three hours.

4 Results

4.1 The panel is competitive, and the families broadly agree

The family-balanced uniform ensemble reaches mean per-assay Spearman $\rho = 0.5330$ across 216 assays; the best single model in the panel (VENUSREM) reaches 0.5345. Both sit in the range of published PROTEINGYM leaderboard values, so the panel is a fair stand-in for the state of the art rather than a weakened strawman. Mean off-diagonal cross-family agreement is $\rho = 0.757$: the families broadly agree, which is precisely why the residual disagreement is interesting—and, as section 4.5 shows, why reweighting them has so little leverage.

4.2 Conflict is strongly structured, but only at the assay level

Assay level. Dataset covariates predict an assay’s mean cross-family disagreement with 5-fold CV $R^2 = 0.414$ (permutation $p = 0.005$; null mean $R^2 = -0.089$); for within-family noise, CV

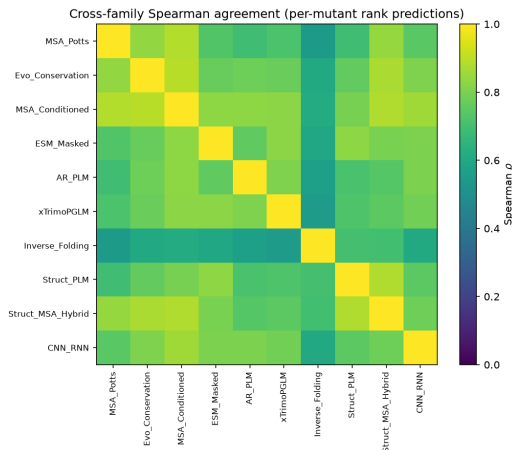


Figure 2: **INVERSE-FOLDING is the most distinctive lineage in the panel.** Cross-family Spearman agreement between per-variant rank predictions, averaged over assays. Mean off-diagonal agreement is 0.757; the inverse-folding row and column are visibly darker, meaning that family agrees least with every other lineage. This single structural fact drives the arbiter’s systematic failure mode in section 4.6.

$R^2 = 0.347$ ($p = 0.005$). Table 2 gives the univariate picture and figure 1 the visual one. Conflict concentrates where the biology is hardest to constrain—shallow alignments, viral proteins, binding assays—and it tracks where the ensemble is wrong (assay EU vs ensemble ρ : -0.338 , permutation $p = 0.0005$). This is the hypothesis’s first claim, and it is comfortably supported.

Mutant level. Within-assay standardised conflict is significantly related to substitution chemistry, but weakly. Mean per-assay Spearman with $z(\text{EU})$ is -0.150 for the BLOSUM62 score (cluster-robust $p = 3.2 \times 10^{-39}$), $+0.074$ for $|\Delta \text{hydrophobicity}|$, and $+0.069$ for $|\Delta \text{volume}|$, while relative position (-0.010) and Δcharge (-0.004) are not significant. A gradient-boosted model over all mutation features, evaluated on *held-out assays*, reaches only $R^2 = 0.054$ (Spearman 0.212, $n = 300,000$). The direction is chemically sensible—the most-contested substitutions are all tryptophan replacements ($W \rightarrow R$ $z = 0.88$, $W \rightarrow A$ 0.84, $W \rightarrow E$ 0.77) and the least-contested are conservative swaps ($I \rightarrow V$ $z = -0.79$, $M \rightarrow L$ -0.72 , $K \rightarrow R$ -0.70)—but it explains a small fraction of the variance. Models agree about conservative substitutions and disagree about the consequences of removing a bulky aromatic, and that is about all the per-variant structure there is.

4.3 Conflict detects failure at the stratum level

Selective prediction. Abstaining on high-conflict variants improves the ensemble’s Spearman on those retained at every coverage level, versus random abstention: $+0.0127$ at 90% coverage ($d = 0.80$), $+0.0254$ at 80% ($d = 1.00$), $+0.0627$ at 50% ($d = 1.26$), and $+0.1222$ at 10% ($d = 1.16$), all with Wilcoxon $p < 10^{-16}$.

The confident-failure test. This is the sharp, falsifiable prediction transplanted from Hamidieh et al. [2026], and it replicates. Splitting each assay at its median AU and median EU gives ensemble accuracy of 0.6350 (low AU, low EU), 0.5267 (low AU, high EU), 0.4337 (high AU, low EU) and 0.3836 (high AU, high EU). Within the low-AU stratum—exactly the variants that a within-paradigm uncertainty estimate would report as confident—high cross-family disagreement costs $\Delta\rho = 0.108$ (Wilcoxon $p = 3.6 \times 10^{-26}$, Cohen’s $d = 0.95$, $n = 214$ assays). Cross-paradigm disagreement carries information that within-paradigm agreement actively hides.

Slice discovery is weak. Clusters fitted on 60% of assays transfer only modestly to the held-out 40% (slice-level conflict transfer $\rho = 0.371$ over 6 slices, which is not a meaningful test at $n = 6$). Two of six slices showed significantly elevated held-out error, notably small or charged residues replaced by bulky aromatics ($G/E/D \rightarrow F/L/W$). We do not claim slice discovery worked.

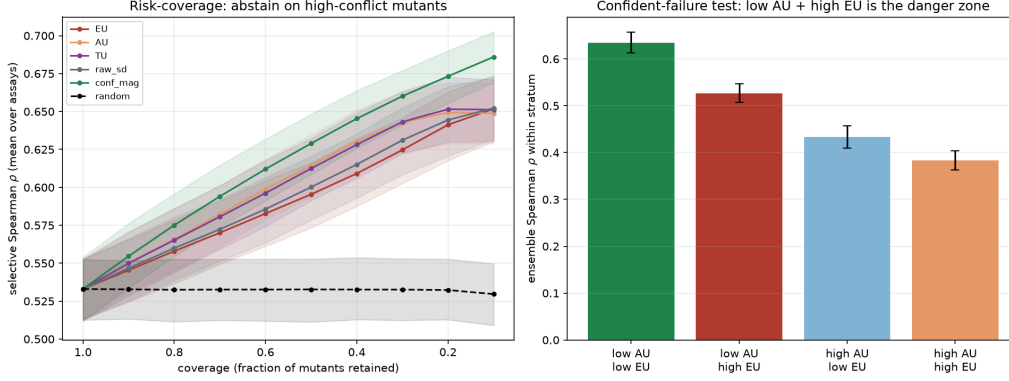


Figure 3: **Abstaining on high-conflict variants improves the retained predictions, and the confident-failure prediction replicates.** (Left) Selective Spearman against coverage; every conflict signal beats random abstention at every coverage level, with the confidence-magnitude control (green) apparently on top—an artefact we dismantle in section 4.4. (Right) Splitting each assay at its median AU and median EU. Within the low-AU stratum—variants a self-consistency estimate would call confident—high cross-family disagreement costs $\Delta\rho = 0.108$ (Wilcoxon $p = 3.6 \times 10^{-26}$, $d = 0.95$).

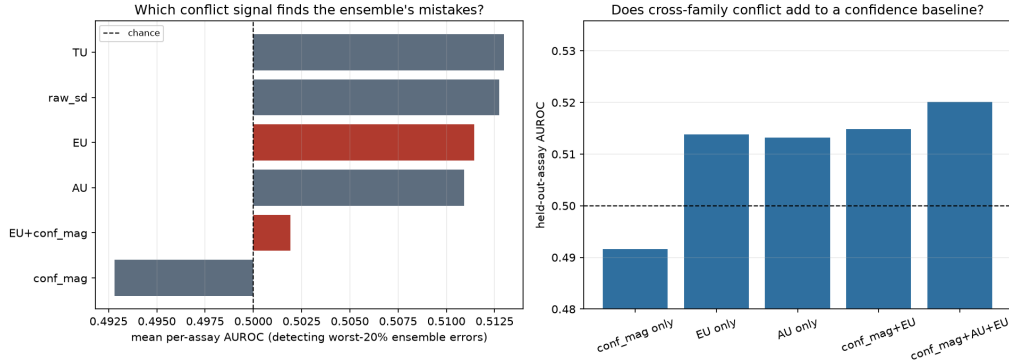


Figure 4: **Under a range-restriction-free metric, the ranking of detectors reverses.** (Left) Mean per-assay AUROC for detecting the worst 20% of ensemble errors. Every conflict signal is above chance; the trivial confidence-magnitude baseline, which appeared to win in figure 3, is below chance. (Right) Held-out assay logistic combination confirms the ordering. Note the absolute scale: an AUROC of 0.51 is a near-useless detector for an individual variant.

4.4 The apparent winner in the selective-prediction curve is a metrics artefact

Ranked by area under the selective-Spearman curve, a trivial confidence-magnitude signal ($|\hat{y} - 0.5|$) beat every conflict signal (0.617 vs EU 0.591). That comparison is an artefact: abstaining on mid-range predictions restricts the retained predictor's range and inflates rank correlation regardless of whether the abstained points were wrong. Re-run as per-assay AUROC for detecting the worst 20% of ensemble errors, the ranking reverses (figure 4): TU 0.5130, raw unbalanced SD 0.5127, EU 0.5114, AU 0.5109, and confidence magnitude 0.4928—at or below chance. EU beats the confidence baseline by +0.019 AUROC ($d = 0.29$, $p = 4.4 \times 10^{-4}$), and held-out logistic combination agrees (conf. magnitude 0.492, EU 0.514, all three signals 0.520).

The honest summary is two-sided. Conflict genuinely orders variants by risk better than chance and better than a confidence baseline, and the widely-used selective-Spearman metric is confounded in a way that hides this. But an AUROC of 0.51 is a near-useless detector for an individual variant: all of the exploitable value sits in aggregate strata, exactly as section 4.2 predicted.

	METHOD	ρ	ρ_{PG}	Δ vs UNIFORM	Cohen's d	Wilcoxon p	Holm p
cheating	ORACLE (weights fitted on this assay's truth)	0.5882	0.5887	+0.0553	1.21	$< 10^{-16}$	$< 10^{-16}$
non-LLM	LEARNED-META (held-out assays)	0.5666	0.5669	+0.0336	0.84	$< 10^{-16}$	$< 10^{-16}$
	LEARNED-META, no conflict features	0.5653	0.5658	+0.0323	0.76	$< 10^{-16}$	$< 10^{-16}$
	LEARNED-META, conflict features only	0.5639	0.5639	+0.0309	0.82	$< 10^{-16}$	$< 10^{-16}$
LLM	CONFLICT-CoT · QWEN3-8B	0.5366	0.5358	+0.0036	0.22	0.0009	0.0093
	DIRECT-ONLY · QWEN3-8B	0.5361	0.5356	+0.0032	0.47	$< 10^{-16}$	$< 10^{-16}$
	CONFLICT-CoT-SHUF · QWEN3-14B (order-randomised)	0.5361	0.5354	+0.0032	0.22	0.0002	0.0021
	CONFLICT-DIRECT · QWEN3-14B	0.5354	0.5348	+0.0025	0.22	0.0074	0.059
	CoT-ONLY-SHUF · QWEN3-14B (order-randomised)	0.5353	0.5346	+0.0002	0.01	0.0056	0.050
	CONFLICT-CoT · QWEN3-14B, greedy	0.5351	0.5344	+0.0022	0.14	0.11	0.57 (n.s.)
	DIRECT-ONLY · QWEN3-14B	0.5291	0.5281	−0.0039	−0.32	$< 10^{-16}$	$< 10^{-16}$
	CoT-ONLY · QWEN3-14B, greedy	0.5289	0.5281	−0.0041	−0.16	0.56	1.00
	CoT-ONLY · QWEN3-8B	0.5272	0.5261	−0.0057	−0.26	0.0001	0.0016
reference	BEST-SINGLE (VENUSREM)	0.5345	0.5354	+0.0015	0.01	0.86	0.86
	UNIFORM (the bar)	0.5330	0.5323	—	—	—	—
	RANDOM-WEIGHTS (20 draws)	0.5300	0.5293	−0.0030	−0.67	$< 10^{-16}$	$< 10^{-16}$

Table 3: **The arbitration ladder**, mean per-assay Spearman over 216 assays, paired against the uniform family-balanced ensemble. ρ_{PG} is the same quantity under PROTEINGYM’s official bias-corrected aggregation; it moves nothing, which indicates the effects are not driven by over-represented proteins. ORACLE is a cheating upper bound: it fixes the total headroom of per-assay reweighting at +0.055. The best legitimate method is a ridge regression (**bold**), which captures 61% of that headroom, while the flagship LLM condition captures 4% and does not survive Holm correction. Three sampled seeds of CONFLICT-CoT on the 14B backbone give $\rho = 0.5351/0.5352/0.5355$, so none of these differences are seed noise.

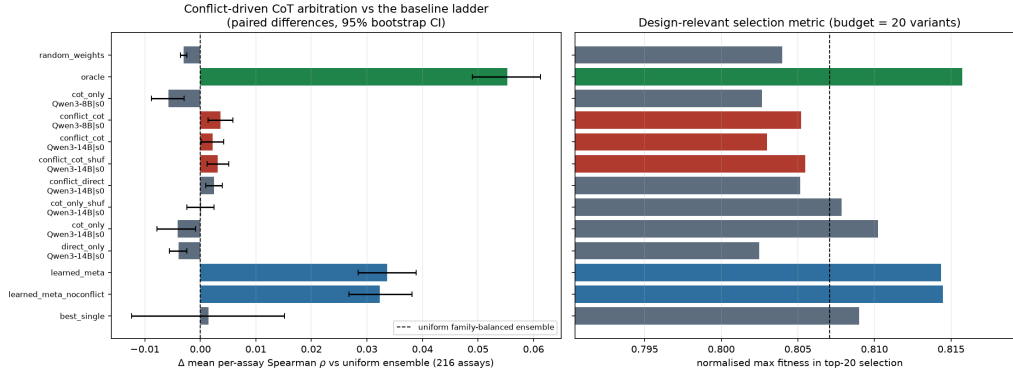


Figure 5: **The full baseline ladder.** (Left) Paired difference in mean per-assay Spearman against the uniform family-balanced ensemble, with 95% bootstrap CIs over 216 assays. The cheating oracle (green) bounds the headroom of per-assay reweighting at +0.055; the non-LLM ridge meta-weighter (blue) captures most of it; the LLM conditions (red and grey) cluster tightly around zero. (Right) The same methods on the design-relevant selection metric at a budget of 20 variants.

4.5 Conflict evidence helps the arbiter; chain-of-thought does not; neither beats a ridge regression

Table 3 gives the ladder and table 4 the ablation contrasts. Four readings follow.

The conflict information helps on the end metric, but the effect is fragile. Against the compute-matched CoT control with the family list in its fixed order, showing the conflict profile improves arbitration on both backbones (+0.0062, $p = 0.0058$ on 14B; +0.0094, $p < 10^{-7}$ on 8B). Under per-assay order randomisation the same contrast falls to +0.0030 and is no longer significant ($d = 0.16$, $p = 0.32$). The reason is visible in section 4.8: with a fixed list, the no-conflict control collapses onto the ordering heuristic—its last-listed family receives a mean weight of 1.36/10—and randomising the order rescues it. A substantial share of the apparent benefit of surfacing conflict was therefore the conflict evidence repairing an artefact in the control, not adding value over a well-behaved control. We report the fixed-order number because it is the design we pre-registered and the randomised number because it is the honest one.

The chain-of-thought contributes nothing. Given the *same* conflict information, reasoning step by step before answering is indistinguishable from answering directly ($\Delta\rho = -0.0003$, $p = 0.49$), and CONFLICT-DIRECT is the *better* of the two on the reasoning-quality metric of section 4.6. CoT without conflict information is actively harmful on the 8B backbone (-0.0089 , $p < 10^{-9}$).

CONTRAST	$\Delta\rho$	Cohen's d	p	READING
Surfacing conflict (CONFLICT-CoT – CoT-ONLY), 14B	+0.0062	0.21	0.0058	evidence helps
Surfacing conflict, 8B	+0.0094	0.32	$< 10^{-7}$	evidence helps
Surfacing conflict, family order randomised, 14B	+0.0030	0.16	0.32 (n.s.)	effect is fragile
Chain-of-thought (CONFLICT-CoT – CONFLICT-DIRECT), 14B	−0.0003	−0.04	0.49 (null)	CoT adds nothing
CoT without conflict (CoT-ONLY – DIRECT-ONLY), 14B	−0.0002	−0.01	0.28	CoT adds nothing
CoT without conflict, 8B	−0.0089	−0.41	$< 10^{-9}$	CoT actively hurts
Order-randomised vs fixed-order CONFLICT-CoT, 14B	+0.0010	0.10	0.0030	randomising is free gain
LLM CONFLICT-CoT vs non-LLM LEARNED-META, 14B	−0.0315	−0.78	$< 10^{-24}$	regression wins
Does conflict add to plain covariates in LEARNED-META?	+0.0013	0.07	0.50 (null)	conflict is redundant

Table 4: **The ablation contrasts, where the interpretation lives.** Paired Wilcoxon over 216 assays (215 for the order-randomised contrast). Showing the conflict profile improves arbitration against a compute-matched CoT control on both backbones, but the effect halves and loses significance once the family list is order-randomised per assay—because randomisation rescues the *control* from a positional pathology (section 4.8). Given the same conflict evidence, reasoning step by step before answering is indistinguishable from answering directly.

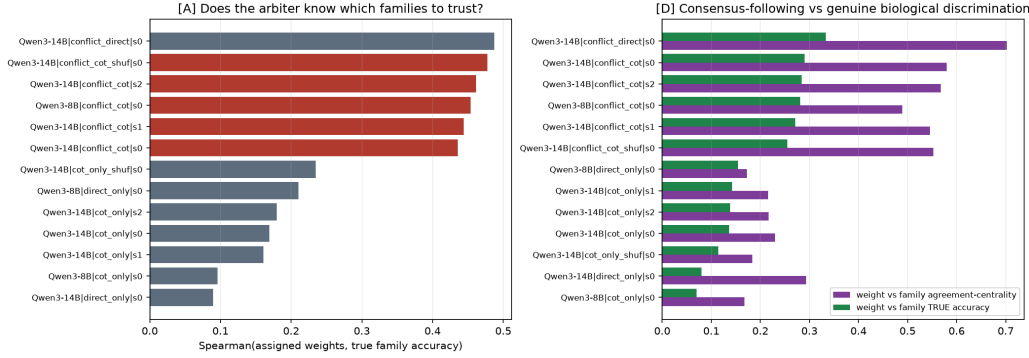


Figure 6: **Surfacing conflict changes what the arbiter knows, but it is still following consensus.** (Left) Spearman correlation between the 10 weights the arbiter assigns and the families’ true per-assay accuracy, which the model never sees. Every conflict-shown condition (red) separates cleanly from every conflict-hidden condition (grey), including under order randomisation. (Right) The same weights correlate more strongly with a family’s agreement-centrality (purple) than with its true accuracy (green), for every condition: the arbiter is executing “trust whoever agrees with the others”.

The hypothesis’s specific claim about chain-of-thought as the instrument is not supported: it is the evidence, not the reasoning format, that does the work.

A ridge regression beats the LLM by an order of magnitude, and does not need the conflict profile. LEARNED-META gains +0.0336—61% of the entire oracle headroom—versus +0.0022 for the flagship LLM condition and +0.0036 for the best LLM condition of any kind ($\Delta = -0.0315$, $d = -0.78$, $p < 10^{-24}$). Its ablations are the more sobering result: covariates *without* any conflict features already deliver +0.0323, conflict features *alone* deliver +0.0309, and adding the conflict profile on top of plain covariates buys +0.0013, which is not significant ($d = 0.07$, $p = 0.50$). The conflict profile is genuinely informative about which families to trust; it is simply not *additional* to what alignment depth, taxon and assay type already say. This is the hypothesis’s own claim taken to its logical conclusion: if conflict clusters around dataset biases, then conditioning on the biases already captures it.

The gains land where the theory says they should. Per-assay improvement over uniform correlates positively with that assay’s conflict level for every conflict-conditioned method (CONFLICT-DIRECT $\rho = 0.379$, CONFLICT-CoT-8B 0.298, CONFLICT-CoT-14B 0.196, LEARNED-META 0.330, all $p < 0.005$) and *negatively* for every method that cannot see the conflict (CoT-ONLY-14B −0.259, DIRECT-ONLY-14B −0.205). Targeting works; there is just very little to win.

4.6 The reasoning is largely correct, and its errors are systematic

Surfacing conflict transforms the arbiter’s knowledge of which families to trust, and this effect survives order randomisation. Correlating the assigned weights against the families’ true

FAMILY	LLM weight rank	True accuracy rank	Centrality rank	Mis-rank
INVERSE-FOLDING	8.94	6.54	9.51	+2.41
CNN_RNN	8.19	6.91	5.32	+1.28
Struct_PLM	4.22	3.11	4.62	+1.11
Evo_Conservation	5.41	4.36	5.26	+1.04
Struct_MSA_Hybrid	3.41	2.48	3.32	+0.93
AR_PLM	7.19	7.54	7.36	−0.35
xTrimoPGLM	6.49	7.42	7.17	−0.93
ESM_Masked	4.23	5.35	3.48	−1.12
MSA-POTTS	4.77	6.73	7.32	−1.96
MSA-CONDITIONED	2.15	4.57	1.65	−2.42

Table 5: **The arbiter tracks agreement-centrality rather than accuracy** (QWEN3-14B, CONFLICT-CoT; 1 = most trusted / most accurate / most central). Positive mis-rank means the family is trusted *less* than its true accuracy deserves; all $p < 10^{-16}$. The two extremes are the two extremes of the centrality column: the least-central family (INVERSE-FOLDING) is the most under-trusted and the most-central family (MSA-CONDITIONED) is the most over-trusted. Inverse-folding models score sequences given the 3D backbone and are *supposed* to disagree with alignment-based models; on the 66 stability assays they achieve the third-highest true Spearman (0.618) yet receive a mean weight of only 4.47/10, while MSA-CONDITIONED receives 9.00/10 for a true Spearman of 0.527.

per-assay Spearman gives 0.487 for CONFLICT-DIRECT (14B; 93.5% of assays positive), 0.477 for CONFLICT-CoT-SHUF, 0.436/0.444/0.461 for the three seeds of CONFLICT-CoT (14B), and 0.454 for CONFLICT-CoT (8B)—against 0.235 for CoT-ONLY-SHUF, 0.169/0.161/0.180 for CoT-ONLY (14B), 0.096 for CoT-ONLY (8B) and 0.090 for DIRECT-ONLY (14B). Showing the conflict roughly doubles weight quality under the clean order-randomised design (0.235 $\rightarrow$ 0.477) and more than doubles it under the fixed-order design (0.169 $\rightarrow$ 0.436 on 14B; 0.096 $\rightarrow$ 0.454 on 8B). Unlike the end-metric effect, this is robust to the ordering confound, stable across three sampled seeds, and consistent across both backbones.

This is the central dissociation of the paper: the same intervention that roughly doubles how well the arbiter knows which models to trust moves the end metric by +0.003 to +0.006—and its best variant on this reasoning metric, CONFLICT-DIRECT (0.487), does no chain-of-thought at all.

The rationales name the right biological factors. Regex probes on the generated text, scored as AUROC for detecting whether that factor is actually present in the assay metadata, give 0.98–1.00 for “viral protein”, 0.97–0.98 for organismal-fitness assays, 0.91–0.92 for binding assays and 0.82–0.89 for stability assays across three seeds of 14B CONFLICT-CoT. The rationales are grounded in the specific assay rather than generic boilerplate. The model is much worse at noticing sequence length (0.54–0.60) and essentially blind to the presence of multi-mutants (0.50–0.51).

But it over-applies the alignment-depth heuristic. The empirically correct relationship between alignment depth and the accuracy of MSA-based families is $\rho = 0.19$. The arbiter’s weights track alignment depth at $\rho = 0.43$ (14B CONFLICT-DIRECT) up to $\rho = 0.87$ (8B CoT-ONLY)—a two- to four-fold over-application of a real but weak rule. The over-application is systematically *smallest* in the conflict conditions (0.42–0.55) and largest without them (0.62–0.87): seeing the actual disagreement data moderates the model’s prior toward the truth.

Much of the apparent skill is consensus-following. For 14B CONFLICT-CoT, assigned weight correlates with a family’s agreement-centrality at $\rho = 0.58$ but with its true accuracy at only $\rho = 0.29$, while centrality itself predicts true accuracy at $\rho = 0.45$. For CONFLICT-DIRECT the pattern is starker (0.70 vs 0.33). The arbiter is executing “trust the family that agrees with the others”—a valid heuristic, executed *worse* than the raw centrality number it was handed. This explains the ridge regression’s advantage directly: a regression on the same numbers does better because it does not add noise on top of them.

The systematic failure mode: outlier $\neq$ wrong. Inspecting rationales revealed a recurring error. On CAPSD_AAV2S_Sinai_2021 (AAV2 capsid, $N_{\text{eff}}/L = 0.26$, viral, organismal fitness) the arbiter writes that “INVERSE-FOLDING’s low agreement with others and high deviation from consensus suggests it may be less reliable here” and assigns it 2/10—while its true Spearman on that assay is 0.459, second best of the ten families (MSA-POTTS, which receives 9/10, achieves 0.350). On ARGR_ECOLI_Tsuboyama_2023_1A0Y (stability, $N_{\text{eff}}/L = 54$) the same reasoning as-

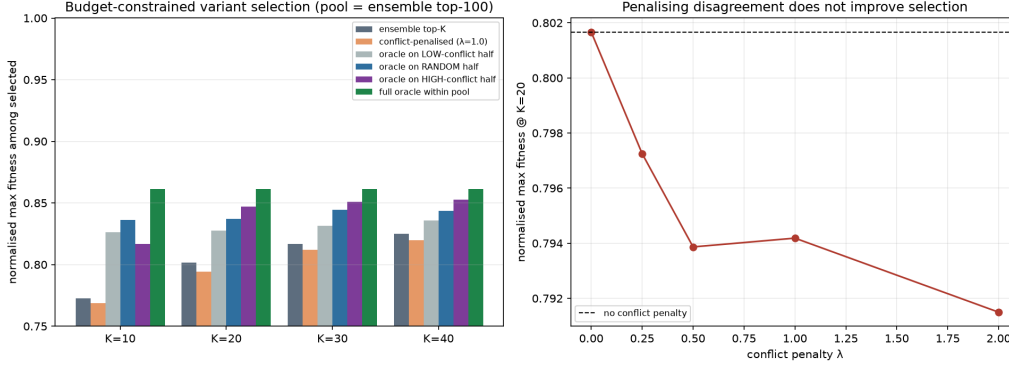


Figure 7: **Budget-constrained variant selection.** (Left) Normalised maximum true fitness among the K selected variants, for budgets $K \in \{10, 20, 30, 40\}$. The partial oracle restricted to the *high*-conflict half of the pool (purple) beats the random half (blue) and the low-conflict half (grey) at $K \geq 20$. (Right) Penalising high-conflict candidates monotonically degrades selection.

METHOD ($K = 20$)	Max fitness @ 20	Δ vs top- K	p	Top-1% hit rate
ensemble top- K (baseline)	0.8017	—	—	0.444
conflict-penalised, $\lambda = 0.5$	0.7939	$-0.0078 \downarrow$	0.0005	0.416
conflict-penalised, $\lambda = 1.0$	0.7942	$-0.0075 \downarrow$	0.0014	0.416
conflict-penalised, $\lambda = 2.0$	0.7915	$-0.0102 \downarrow$	0.0001	—
oracle on LOW-conflict half of pool	0.8274	$+0.0258$	$< 10^{-16}$	—
oracle on RANDOM half of pool	0.8372	$+0.0356$	$< 10^{-16}$	0.657
oracle on HIGH-conflict half of pool	0.8472	$+0.0455 \uparrow$	$< 10^{-16}$	0.720
full oracle within pool	0.8615	$+0.0598$	$< 10^{-16}$	0.787

Table 6: **Conflict is a good router and a bad filter.** Budget-constrained selection over 207 assays; the pool is the ensemble’s top-100 variants per assay and the metric is normalised maximum true fitness among the $K = 20$ selected (0 = worst variant in the assay, 1 = best). Penalising high-conflict candidates makes selection significantly *worse* at every λ . But perfectly resolving only the high-conflict half of the pool recovers 76.1% of the full-oracle gain, versus 59.4% for a random half and 43.0% for the low-conflict half: conflict identifies where the fixable errors are, even though it cannot say which way to fix them.

signs INVERSE-FOLDING 2/10 against a true Spearman of 0.733, by far the best of the ten (the family weighted 10/10 achieves 0.435).

This is not anecdotal. Table 5 ranks the families per assay by assigned weight, by true accuracy and by agreement-centrality, and the mis-ranking is monotone in centrality. Inverse-folding models score sequences given the 3D backbone; they are supposed to disagree with alignment-based models, and on stability assays that disagreement is them being right. The arbiter conflates “this family is an outlier” with “this family is wrong”—exactly the failure the agreement matrix in figure 2 predicts, since INVERSE-FOLDING is the most distinctive lineage in the panel.

4.7 Conflict is a good router and a bad filter

Two clean and opposite conclusions come out of table 6. *As a filter, conflict fails:* preferring low-conflict variants significantly reduces selection quality at every λ and every budget, and the variants that were picked and turned out poor are not more conflicted than those picked and good ($\Delta z = -0.002$, $p = 0.63$)—the same null as the AUROC of 0.51 in section 4.4, seen from the design side. *As a router, conflict works:* perfectly resolving only the high-conflict half of the pool recovers 76.1% of the full-oracle gain at $K = 20$, versus 59.4% for a random half and 43.0% for the low-conflict half ($K = 30$: 76.5% / 62.1% / 33.0%; $K = 40$: 76.0% / 51.4% / 29.5%). The ordering reverses at $K = 10$ (49.7% / 71.6% / 60.6%), where the top-10 picks are dominated by high-confidence low-conflict variants. Conflict correctly identifies where the fixable errors are even though it cannot say which way to fix them.

	CLAIM	VERDICT	EVIDENCE
C1	Conflict clusters around biological ambiguities and dataset biases	supported at assay level, weak at variant level	CV $R^2 = 0.414$ ($p = 0.005$); MSA depth $\rho = -0.43$; virus +27% over human; but held-out mutation-level $R^2 = 0.054$
C2	CoT is the right instrument for surfacing conflict	split : surfacing supported for reasoning quality, CoT refuted	Weight quality $0.235 \rightarrow 0.477$ under order randomisation (both backbones, 3 seeds); end metric +0.003 ($p = 0.32$) once order is randomised; CoT vs direct with identical information -0.0003 ($p = 0.49$)
C3	Conflict detects failure modes	supported at stratum level, refuted at variant level	Confident-failure $\Delta\rho = 0.108$ ($d = 0.95$); selective abstention beats random at all coverages; but per-variant error AUROC = 0.511
C4	Targeted consensus improves design	not supported	CONFLICT-CoT +0.0022, Holm $p = 0.57$; conflict-penalised selection significantly worse; a non-LLM ridge baseline gains an order of magnitude more

Table 7: Verdicts on the four decomposed claims, against pre-registered success criteria.

This also bounds the whole enterprise: the maximum any per-variant conflict arbiter could deliver at $K = 20$ is +0.046 normalised max fitness, and it would have to be a near-perfect arbiter.

4.8 Robustness: a positional confound was doing part of the work

Positional bias is real, large, and it changes one conclusion. The family list order was fixed across assays in the primary design. Assigned weight correlates with list position at Spearman -0.22 to -0.47 (earlier-listed families get higher weights, all $p < 10^{-16}$). The bias is weakest in the conflict conditions (14B CONFLICT-CoT: -0.238) and strongest without conflict information (8B CoT-ONLY: -0.474 ; 14B DIRECT-ONLY: -0.429). The extreme case is diagnostic: under fixed order, 14B CoT-ONLY assigns the first-listed family a mean weight of 7.56/10 and the last-listed 1.36/10—a near-total collapse onto position.

Randomising the family list order per assay. Under randomisation the positional slope weakens (CONFLICT-CoT-SHUF -0.189 , CoT-ONLY-SHUF -0.160) and the last-listed family’s mean weight rises to 5.53 and 5.13. The consequences are asymmetric. The reasoning-quality effect survives cleanly (0.477 vs 0.235). The end-metric effect does not: the surfacing contrast drops from +0.0062 ($p = 0.0058$) to +0.0030 ($p = 0.32$), because the randomised control improves far more than the randomised treatment does. Separately, CONFLICT-CoT-SHUF is the better configuration overall (+0.0032 vs uniform, Holm $p = 0.0021$, against +0.0022 and Holm $p = 0.57$ for fixed order; direct contrast +0.0010, $p = 0.0030$). Randomising the list is a free improvement.

Backbone agreement rises sharply with conflict information. Within-assay Spearman between QWEN3-14B and QWEN3-8B weight vectors is 0.780 for CONFLICT-CoT, 0.540 for CoT-ONLY and 0.498 for DIRECT-ONLY. Showing the conflict anchors the two backbones onto the same answer; without it, arbitration is substantially backbone-specific.

Sampling stochasticity is moderate, and lower with conflict information. Between two temperature-0.7 seeds, within-assay weight agreement is $\rho = 0.871$ for CONFLICT-CoT and 0.697 for CoT-ONLY. The three CONFLICT-CoT seeds give $\rho = 0.5351/0.5352/0.5355$, so none of the reported effects are seed artefacts.

No simple length confound. Weight dispersion does not track prompt length monotonically: CONFLICT-DIRECT has the second-longest prompt and the lowest weight dispersion (1.70), while CoT-ONLY has a short prompt and high dispersion (2.55).

5 Discussion

5.1 Verdict on each claim

We decomposed the hypothesis into four separately falsifiable claims before running anything, with pre-registered success criteria. Table 7 gives the verdicts.

The criteria are met for C1 (CV $R^2 = 0.414 > 0.15$, $p = 0.005$) and C3 (both conditions met), partially met for C2—the conflict-shown conditions beat the compute-matched control robustly on

reasoning quality and only under the fixed-order design on the end metric, and the CoT component specifically does not help at all—and not met for C4.

5.2 The central tension, and what explains it

The most informative result is a dissociation. The intervention that roughly doubles how well the arbiter knows which models to trust ($0.235 \rightarrow 0.477$ weight–accuracy correlation under the clean order-randomised design) moves the end metric by $+0.003$, which is not significant. Why?

Because **a family-balanced 10-member ensemble is extremely robust to reweighting**. Mean cross-family agreement is $\rho = 0.757$, so the families are largely making the same prediction; even the oracle, which knows each family’s true accuracy on the assay it is weighting, gains only $+0.055$. There is only 0.055 of headroom in the entire reweighting operation. The LLM captures 4–7% of it; a ridge regression captures 61%. The bottleneck is not the arbiter’s knowledge—it is that per-assay reweighting is a low-leverage lever.

A second cause is visible in the consensus-following analysis: knowing the *ordering* of family reliability is not the same as knowing the right *magnitudes*. The arbiter emits integers 0–10 with a mean within-assay dispersion of about 2.2, so even a correctly-ordered weight vector applies a much blunter correction than the ridge model’s calibrated one. The productive direction is therefore not better arbitration of weights but arbitration that can change *rankings within* an assay, which has roughly ten times the headroom ($+0.046$ on max fitness at $K = 20$ from resolving only the high-conflict half of the pool)—while section 4.4 shows we currently have no signal that says which way to resolve them.

5.3 Relation to prior work

We **confirm** the core prediction of Hamidieh et al. [2026] in a new domain: epistemic disagreement flags confident failures that aleatoric agreement misses, with a large effect ($d = 0.95$). Their formalism transplants cleanly from LLM self-consistency to research-lineage families. We also **confirm** the debate and CoT correctives [Smit et al., 2023, Zhang et al., 2025b]: once test-time compute is matched, the reasoning format contributes nothing here, and CoT without useful evidence is actively harmful on the smaller backbone. Any protein–LLM paper reporting a CoT gain without a compute-matched control should be read with this in mind. We **extend** Tan et al. [2026] by testing precisely the conditioning they omit, and find that conditioning on conflict does help relative to not conditioning, but that the residual headroom is far smaller than it looks and is more efficiently captured by a regression than by an LLM; their own observation that the LLM backbone barely affects results is consistent with our finding that end-task performance is nearly insensitive to arbitration quality. Finally, we **add** a methodological caution absent from the fitness-prediction literature: the selective-Spearman metric widely used to demonstrate uncertainty quantification is confounded by range restriction, and a trivial confidence-magnitude baseline appears to win under it while performing at chance under a clean metric.

5.4 Surprises worth reporting

The metric artefact. Our first pass concluded that conflict was *worse* than a trivial confidence baseline for abstention. That conclusion was an artefact of the metric and it reversed under a range-restriction-free measure. We report both, because the artefact is a trap the field is currently walking into.

Outlier $\neq$ wrong. The arbiter’s most consequential error is structural, not random: it reads “deviates from consensus” as “unreliable” and therefore systematically penalises inverse-folding models, which are the most distinctive lineage in the panel and are often the *most* accurate—dramatically so on stability assays ($\rho = 0.733$ vs 0.435 for the family it preferred). Any consensus mechanism built on agreement-centrality inherits this bias. Fixing it likely requires telling the arbiter *why* a family is expected to deviate, not just that it does.

Conflict is redundant with plain metadata. Conflict features alone ($+0.0309$) and covariates alone ($+0.0323$) reach almost the same place, and combining them adds $+0.0013$ ($p = 0.50$). Disagreement is largely a *readout* of alignment depth and assay type rather than an independent signal—a direct, quantitative version of the hypothesis’s own claim that conflicts cluster around

dataset biases, taken to its logical conclusion: if conflict is fully explained by the biases, it is not extra information.

A positional confound was doing part of the work. The order-randomised control, added after robustness testing revealed the bias, cut the headline surfacing effect roughly in half and removed its significance. The mechanism is instructive: the *control* condition was the one damaged by fixed ordering, so the conflict evidence was partly being credited for repairing an artefact. This is a general hazard for any LLM-as-judge design that presents a fixed list of options—list-order randomisation should be the default, not a robustness check.

5.5 Limitations

Methodological. (i) The primary design used a fixed family order; the order-randomised control was run to address this and it *changed a conclusion*, but it covers only the 14B backbone at one seed for two conditions, so the randomised estimates are less well powered than the fixed-order ones. A pre-registered design would have randomised from the start for every condition. (ii) The panel is one choice among many: family boundaries and the choice of 3 members per family are defensible but not unique, and all cross-family conflict results are conditional on this grouping. (iii) The LEARNED-META softmax temperature was fixed a priori rather than tuned; a tuned version would presumably widen its lead, so this does not threaten the direction of the comparison. (iv) We arbitrate at the assay level, where E1 and E2 said the signal was; a variant-level arbiter was not run, and while section 4.4 and section 4.7 bound its achievable gain, “we did not try it” is not the same as “it cannot work”. (v) Only local open-weight models were available. QWEN3-14B is not a frontier model, though the backbone-invariance reported by Tan et al. [2026] and our own 8B-versus-14B comparison (8B slightly *better* on the primary contrast) argue against a large backbone effect.

Data and scope. Contamination cannot be excluded: Qwen3 was trained on web text that includes PROTEINGYM documentation. It is implausible that it memorised per-assay per-family Spearman values at the required granularity, and the conflict-hidden conditions—which give protein identity without conflict data—perform *worse* than uniform, which argues against memorised per-assay knowledge; but we cannot prove absence. PROTEINGYM assays are also not a random sample of protein engineering problems: they are biased toward proteins with deep alignments and cheap assay types [Shaw et al., 2026], so our covariate conclusions are conditional on that distribution. All claims are computational; the design-selection metric is a retrospective proxy over already-measured variants, not a prospective test. And the hypothesis’s literal domain—genomic foundation models—is not tested here.

Threats to validity. The ladder is Holm-corrected and covariate tests are FDR-corrected, but the study as a whole ran many analyses, and borderline results (CONFLICT-DIRECT, Holm $p = 0.059$; CoT-ONLY-SHUF, Holm $p = 0.050$) should be treated as suggestive. The Holm correction is applied across a ladder that grew as the study progressed, which makes it conservative for later-added conditions—another reason to read the LLM effects as small rather than as absent-or-present. Rationale probes are regexes rather than semantic classifiers, so mention rates are lower bounds. Most importantly, the LLM effects (+0.002 to +0.009 Spearman) are statistically detectable only because $n = 216$ paired assays; they are of no practical significance, and we do not claim otherwise.

5.6 Broader implications

Three recommendations follow for practitioners. First, report and use cross-family disagreement as an *assay-level* trust score: it is cheap, it is strongly predicted by alignment depth and taxon, and it correctly anticipates where an ensemble will do poorly. Second, do not use per-variant disagreement to filter candidate variants—penalising high-conflict variants makes budget-constrained selection significantly worse. Use it as a *router* instead, sending the high-conflict half of a candidate pool to whatever more expensive resolution is available. Third, if you are going to spend an LLM on ensemble weighting, spend it on the evidence rather than the reasoning format: showing the conflict numbers is what changes the arbiter’s beliefs, chain-of-thought over the same numbers adds nothing measurable, and the option list must be order-randomised or a positional artefact will masquerade as a treatment effect.

There is also a warning for evaluation practice beyond this domain. We measured reasoning quality and end-task performance separately and found them sharply dissociated: an intervention that dou-

bles the arbiter’s knowledge of which models to trust moves the score by less than half a percent. LLM-arbiter evaluations that report only end metrics are systematically blind to whether the reasoning improved at all—and, symmetrically, evaluations that report only rationale quality can claim a large effect on a task where it cannot possibly pay.

6 Conclusion

Cross-model conflict in protein fitness prediction *is* systematically structured around biological ambiguities and dataset biases—decisively so at the level of assays and strata, where dataset covariates explain 41% of its out-of-sample variance, shallow alignments and viral proteins carry the most conflict, and cross-paradigm disagreement flags confident ensemble failures that within-paradigm agreement conceals ($d = 0.95$). It is *not* meaningfully structured at the level of individual variants, where every conflict signal detects errors at barely above chance (AUROC = 0.511). Surfacing that conflict to an LLM makes its arbitration markedly better informed—roughly doubling the correlation between assigned weights and true model accuracy, robustly across backbones, seeds, and an order-randomised design—but chain-of-thought reasoning over the same information adds nothing, the end-task improvement is negligible and loses significance once the ordering confound is removed, and a simple regression given the same inputs captures an order of magnitude more of the available headroom.

Key takeaway. The hypothesis is substantially confirmed for detection and refuted for improvement, with the specific mechanism it proposes—chain-of-thought—shown to be the inactive ingredient. It is the conflict *evidence*, not the reasoning format, that changes what the arbiter knows; and even perfect knowledge of which model family to trust is worth little when the families agree at $\rho = 0.757$.

Future work. Four directions follow directly. (i) Attack the headroom that exists: within-pool re-ranking has roughly ten times the leverage of per-assay reweighting, and the open question is whether any signal—structural context such as solvent accessibility, contact number, or proximity to the active site, none of which is in our feature set—can say which side of a variant-level conflict is right. (ii) Fix the outlier-is-wrong bias by giving the arbiter per-family *expected* deviation profiles, and test whether the inverse-folding mis-ranking disappears and weight quality exceeds agreement-centrality. (iii) Test the assay-level trust score prospectively: predict the conflict of newly published assays from metadata alone and check that predicted conflict anticipates ensemble accuracy out of sample and out of time. (iv) Replicate on genomic foundation models once a genomic benchmark publishes a comparable cross-model per-example score matrix, so the hypothesis can be tested in its literal domain.

References

- Muxi Chen, Chenchen Gao, and Qiang Xu. HiBug2: Efficient and interpretable error slice discovery for comprehensive model debugging. *arXiv preprint arXiv:2501.16751*, 2025.
- Hugo Dalla-Torre, Guillaume Richard, and Thomas Pierrot. GENEB: Why genomic models are hard to compare. *arXiv preprint arXiv:2606.04525*, 2026.
- Justas Dauparas, Ivan Anishchenko, Nathaniel Bennett, Hua Bai, Robert J Ragotte, Lukas F Milles, Basile I M Wicky, Alexis Courbet, Rob J de Haas, Neville Bethel, et al. Robust deep learning-based protein sequence design using ProteinMPNN. *Science*, 378(6615):49–56, 2022.
- Sabri Eyuboglu, Maya Varma, Khaled Saab, Jean-Benoit Delbrouck, Christopher Lee-Messer, Jared Dunnmon, James Zou, and Christopher Ré. Domino: Discovering systematic errors with cross-modal embeddings. In *International Conference on Learning Representations (ICLR)*, 2022.
- Jonathan Frazer, Pascal Notin, Mafalda Dias, Aidan Gomez, Joseph K Min, Kelly Brock, Yarin Gal, and Debora S Marks. Disease variant prediction with deep generative models of evolutionary data. *Nature*, 599:91–95, 2021.
- Kimia Hamidieh, Veronika Thost, Walter Gerych, Mikhail Yurochkin, and Marzyeh Ghassemi. Complementing self-consistency with cross-model disagreement for uncertainty quantification. In *International Conference on Learning Representations (ICLR)*, 2026.

- Thomas A Hopf, John B Ingraham, Frank J Poelwijk, Charlotta P I Schärfe, Michael Springer, Chris Sander, and Debora S Marks. Mutation effects predicted from sequence co-variation. *Nature Biotechnology*, 35(2):128–135, 2017.
- Chloe Hsu, Robert Verkuil, Jason Liu, Zeming Lin, Brian Hie, Tom Sercu, Adam Lerer, and Alexander Rives. Learning inverse folding from millions of predicted structures. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2022.
- Kaiyi Jiang, Yiming Zhu, and James J Collins. Inference-only dropout improves zero-shot protein fitness prediction with protein language models. *arXiv preprint arXiv:2506.14793*, 2025.
- Elodie Laine, Yasaman Karami, and Alessandra Carbone. GEMME: A simple and fast global epistatic model predicting mutational effects. *Molecular Biology and Evolution*, 36(11):2604–2619, 2019.
- Jingwei Lee, Wenxuan Zhou, and Muhao Chen. Can reasoning help large language models capture human annotator disagreement? *arXiv preprint arXiv:2506.19467*, 2025.
- Mingchen Li, Yang Tan, Xinzhu Ma, Bozitao Zhong, Huiqun Yu, Ziyi Zhou, Wanli Ouyang, Bingxin Zhou, Liang Hong, and Pan Tan. ProSST: Protein language modeling with quantized structure and disentangled attention. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023.
- Joshua Meier, Roshan Rao, Robert Verkuil, Jason Liu, Tom Sercu, and Alexander Rives. Language models enable zero-shot prediction of the effects of mutations on protein function. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- Erik Nijkamp, Jeffrey A Ruffolo, Eli N Weinstein, Nikhil Naik, and Ali Madani. ProGen2: Exploring the boundaries of protein language models. *Cell Systems*, 14(11):968–978, 2023.
- Pascal Notin, Mafalda Dias, Jonathan Frazer, Javier Marchena-Hurtado, Aidan Gomez, Debora S Marks, and Yarin Gal. Tranception: Protein fitness prediction with autoregressive transformers and inference-time retrieval. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2022.
- Pascal Notin, Aaron W Kollasch, Daniel Ritter, Lood van Niekerk, Steffanie Paul, Hansen Spinner, Nathan Rollins, Ada Shaw, Ruben Weitzman, Jonathan Frazer, Mafalda Dias, Dinko Franceschi, Rose Orenbuch, Yarin Gal, and Debora S Marks. ProteinGym: Large-scale benchmarks for protein fitness prediction and design. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Adam J Riesselman, John B Ingraham, and Debora S Marks. Deep generative models of genetic variation capture the effects of mutations. In *Nature Methods*, volume 15, pages 816–822, 2018.
- Ada Shaw, Aaron W Kollasch, and Debora S Marks. Better protein function prediction by modeling survivorship bias. *arXiv preprint arXiv:2605.06879*, 2026.
- Andries Smit, Paul Duckworth, Nathan Grinsztajn, Thomas D Barrett, and Arnu Pretorius. Should we be going MAD? a look at multi-agent debate strategies for LLMs. *arXiv preprint arXiv:2311.17371*, 2023.
- Jin Su, Chenchen Han, Yuyang Zhou, Junjie Shan, Xibin Zhou, and Fajie Yuan. SaProt: Protein language modeling with structure-aware vocabulary. In *International Conference on Learning Representations (ICLR)*, 2024.
- Yang Tan, Bingxin Yu, Jia Wu, Bozitao Zhong, Mingchen Li, Guisheng Fan, Yanjie Zhu, Huiqun Liang, Liang Dong, and Liang Hong. Rank-and-reason: Multi-agent collaboration accelerates zero-shot protein mutation prediction. *arXiv preprint arXiv:2602.00197*, 2026.

- Timothy F Truong Jr and Tristan Bepler. PoET: A generative model of protein families as sequences-of-sequences. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. Knowledge conflicts for LLMs: A survey. *arXiv preprint arXiv:2403.08319*, 2024.
- Chen Zhang, Sourav Bhattacharya, and Alexander Karollus. Same model, better performance: The impact of shuffling on DNA language model benchmarking. *arXiv preprint arXiv:2510.12617*, 2025a.
- Hangfan Zhang, Zhiyao Cui, Qiaosheng Wang, and Shuyue Hu Li. Stop overvaluing multi-agent debate: Rethink evaluation and embrace model heterogeneity. *arXiv preprint arXiv:2502.08788*, 2025b.
- Yongjin Zhang, Jonas Becker, Jan Philip Kang, Haonan Xia, and Iryna Gurevych. If multi-agent debate is the answer, what is the question? problem drift in long debates. *arXiv preprint arXiv:2502.19559*, 2025c.

A Additional experimental detail

A.1 Selective prediction at every coverage level

Table 8 gives the full risk–coverage table summarised in section 4.3. Abstention is by descending EU; the control abstains uniformly at random on the same number of variants per assay.

Coverage retained	Selective ρ (EU)	Δ vs random	Cohen’s d	Wilcoxon p
90%	0.5454	+0.0127	0.80	$< 10^{-16}$
80%	0.5579	+0.0254	1.00	$< 10^{-16}$
50%	0.5954	+0.0627	1.26	$< 10^{-16}$
20%	0.6413	+0.1089	1.23	$< 10^{-16}$
10%	0.6518	+0.1222	1.16	$< 10^{-16}$

Table 8: Selective Spearman under conflict-based abstention, 216 assays.

A.2 The confident-failure 2×2

Splitting each assay at its median AU and median EU (section 4.3) gives the ensemble accuracies in table 9. The comparison that matters is the first two rows: within the low-AU stratum, where within-paradigm agreement reports confidence, high cross-paradigm disagreement costs $\Delta\rho = 0.108$.

Stratum	Ensemble ρ	95% CI
low AU, low EU	0.6350	[0.613, 0.656]
low AU, high EU	0.5267	[0.505, 0.547]
high AU, low EU	0.4337	[0.410, 0.459]
high AU, high EU	0.3836	[0.363, 0.406]

Table 9: Ensemble accuracy within each AU/EU stratum, bootstrap CIs over assays.

A.3 Per-assay error-detection AUROC

Table 10 gives the range-restriction-free detector comparison of section 4.4: mean per-assay AU-ROC for identifying the worst 20% of ensemble errors. All conflict signals are above chance and the confidence-magnitude baseline is at or below it, reversing the ordering seen under selective Spearman.

Signal	Mean per-assay AUROC	Fraction of assays > 0.5	p
TU	0.5130	0.542	0.024
raw (unbalanced) SD	0.5127	0.519	0.033
EU	0.5114	0.537	0.043
AU	0.5109	0.547	0.022
confidence magnitude	0.4928	0.491	0.49

Table 10: Error-detection AUROC. EU beats the confidence baseline by +0.019 ($d = 0.29$, $p = 4.4 \times 10^{-4}$).

A.4 Weight quality across all run configurations

Table 11 gives the full version of the reasoning-quality result of section 4.6: the Spearman correlation between the 10 weights the arbiter assigns for an assay and the families’ true per-assay accuracy on that assay, which never enters a prompt.

Run	Conflict shown	$\rho(\text{weights, true accuracy})$	Frac. assays positive	p
CONFLICT-DIRECT · 14B	yes	0.487	0.935	3.9×10^{-35}
CONFLICT-CoT-SHUF · 14B	yes	0.477	0.917	4.7×10^{-34}
CONFLICT-CoT · 14B, seed 2	yes	0.461	0.935	3.4×10^{-34}
CONFLICT-CoT · 8B	yes	0.454	0.898	3.2×10^{-33}
CONFLICT-CoT · 14B, seed 1	yes	0.444	0.907	8.7×10^{-33}
CONFLICT-CoT · 14B, greedy	yes	0.436	0.894	8.5×10^{-33}
CoT-ONLY-SHUF · 14B	no	0.235	0.726	1.3×10^{-16}
DIRECT-ONLY · 8B	no	0.210	0.764	8.2×10^{-21}
CoT-ONLY · 14B, seed 2	no	0.180	0.690	1.2×10^{-12}
CoT-ONLY · 14B, greedy	no	0.169	0.727	1.9×10^{-12}
CoT-ONLY · 14B, seed 1	no	0.161	0.722	8.5×10^{-12}
CoT-ONLY · 8B	no	0.096	0.597	5.5×10^{-4}
DIRECT-ONLY · 14B	no	0.090	0.644	1.0×10^{-5}

Table 11: Weight quality for all 13 run configurations, 216 assays each (215 for CoT-ONLY-SHUF, which had one unparsed generation). Every conflict-shown condition separates from every conflict-hidden condition, including under order randomisation.

A.5 Prompt structure

Each prompt names the protein, the assay type, the taxon, the alignment depth and coverage, the sequence length, the number of variants and the fraction of multi-mutants, then lists the 10 model families with a one-line description of each. In the conflict conditions it additionally supplies, per family, that family’s mean agreement with the other nine on this assay, its mean deviation from the family-balanced consensus, and the assay’s mean AU and EU. It then asks for an integer weight 0–10 per family, in a fixed output format; the CoT conditions ask for a short analysis before the weights, the direct conditions ask for the weights alone. No experimental fitness score and no per-assay model accuracy appears anywhere in any prompt, which we verified by keyword search over all 1,296 prompts.

B Reproducibility

All results derive from PROTEINGYM v1.3’s published zero-shot score matrix plus locally-run QWEN3-14B and QWEN3-8B generations; no protein model was run. We fix SEED = 42 throughout. The environment is Python 3.12.8 with numpy 2.2.6, pandas 3.0.5, scipy 1.18.0, scikit-learn 1.9.0, statsmodels 0.14.6, matplotlib 3.11.1, PyTorch 2.13.0+cu130 and transformers 5.15.0, on 2× NVIDIA RTX A6000 (48 GB) and 32 CPU cores, with a total wall-clock of approximately three hours. We retain the full text of all 2,808 generations, all 1,296 prompts, the per-variant conflict panel (2.47M rows), and the per-assay outputs of every experiment.